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\\Research

Investigators - validated on FIT

Frederick

Mair test

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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\Localizer_aligned

TA: 21 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	104.0 ms
TE	3.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	104.0 ms
TE	3.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	15 deg
Fat-Water Contrast	Standard

Contrast - Common

Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	75 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8

Geometry - Common

Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	104.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	2
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	3
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
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System - Adjustments

B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	104.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	75 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off

Inline - Subtraction

Save Original Images	On
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Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	15 deg
Measurements	1
Contrasts	1
TE	3.00 ms
TR	104.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\SpinEchoFieldMap_AP

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0x2.0x2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	8036.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8036.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	On
EPI Factor	104

Sequence - Part 2

Introduction	Off
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Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\SpinEchoFieldMap_PA

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0x2.0x2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	8036.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8036.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	On
EPI Factor	104

Sequence - Part 2

Introduction	Off
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Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T1w_MPR

TA: 6:54 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
TE	2.22 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	2500.0 ms
TE	2.22 ms
Magn. Preparation	Non-sel. IR
TI	1000 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain

Geometry - AutoAlign

Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.2 A13.8 H12.4 mm
! Orientation	T > C-8.9 > S-0.5
! Rotation	4.64 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1000 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.22 ms
TR	2500.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear

Sequence - Part 1

Bandwidth	220 Hz/Px
Echo Spacing	7.54 ms
Asymmetric Echo	Allowed
Turbo Factor	256

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T1w_MPR

TA: 6:54 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
TE	2.22 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	2500.0 ms
TE	2.22 ms
Magn. Preparation	Non-sel. IR
TI	1000 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain

Geometry - AutoAlign

Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	R1.2 A13.8 H12.4 mm
I Orientation	T > C-8.9 > S-0.5
I Rotation	4.64 deg
I A >> P	208 mm
I R >> L	208 mm
I F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1000 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.22 ms
TR	2500.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear

Sequence - Part 1

Bandwidth	220 Hz/Px
Echo Spacing	7.54 ms
Asymmetric Echo	Allowed
Turbo Factor	256

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T2w_SPC

TA: 5:57 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
TE	561.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	3200.0 ms
TE	561.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm

Geometry - AutoAlign

H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.2 A13.8 H12.4 mm
! Orientation	T > C-8.9 > S-0.5
! Rotation	4.64 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	744 Hz/Px
Echo Spacing	3.98 ms
Turbo Factor	314
Echo Train Duration	1182 ms

Sequence - Part 2

Introduction	On
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T2w_SPC

TA: 5:57 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
TE	561.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	3200.0 ms
TE	561.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm

Geometry - AutoAlign

H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.2 A13.8 H12.4 mm
! Orientation	T > C-8.9 > S-0.5
! Rotation	4.64 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	744 Hz/Px
Echo Spacing	3.98 ms
Turbo Factor	314
Echo Train Duration	1182 ms

Sequence - Part 2

Introduction	On
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\rfMRI_REST_AP

TA: 6:40 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
TE	37.00 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain

Contrast - Common

TR	800.0 ms
TE	37.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	42 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	488
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	Ext./Trigger
Average Cycle	No Signal ms
Acquisition Window	800 ms
Trigger Pulse	1
Trigger Delay	0 ms
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	8
Phases	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	488
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	6600 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\rfMRI_REST_AP

TA: 6:40 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
TE	37.00 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain

Contrast - Common

TR	800.0 ms
TE	37.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	42 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	488
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	Ext./Trigger
Average Cycle	No Signal ms
Acquisition Window	800 ms
Trigger Pulse	1
Trigger Delay	0 ms
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	8
Phases	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	488
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	6600 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCD_dMRI_DistortMAP_AP

TA: 30 sec Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	3
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	50 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off
Disable freq. update	Off

Sequence - Special

Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCD_dMRI_DistortMAP_AP

TA: 30 sec Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	3
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	50 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off
Disable freq. update	Off

Sequence - Special

Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\rfMRI_REST_AP

TA: 6:40 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
TE	37.00 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain

Contrast - Common

TR	800.0 ms
TE	37.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	42 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	488
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	Ext./Trigger
Average Cycle	No Signal ms
Acquisition Window	800 ms
Trigger Pulse	1
Trigger Delay	0 ms
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	8
Phases	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	488
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	6600 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T1w_MPR

TA: 6:54 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
TE	2.22 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	2500.0 ms
TE	2.22 ms
Magn. Preparation	Non-sel. IR
TI	1000 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain

Geometry - AutoAlign

Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	R1.2 A13.8 H12.4 mm
I Orientation	T > C-8.9 > S-0.5
I Rotation	4.64 deg
I A >> P	208 mm
I R >> L	208 mm
I F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1000 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.22 ms
TR	2500.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear

Sequence - Part 1

Bandwidth	220 Hz/Px
Echo Spacing	7.54 ms
Asymmetric Echo	Allowed
Turbo Factor	256

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T1w_MPR

TA: 6:54 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
TE	2.22 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	2500.0 ms
TE	2.22 ms
Magn. Preparation	Non-sel. IR
TI	1000 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain

Geometry - AutoAlign

Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.2 A13.8 H12.4 mm
! Orientation	T > C-8.9 > S-0.5
! Rotation	4.64 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1000 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.22 ms
TR	2500.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear

Sequence - Part 1

Bandwidth	220 Hz/Px
Echo Spacing	7.54 ms
Asymmetric Echo	Allowed
Turbo Factor	256

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\t2_space_sag_cs4_iso

TA: 2:22 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 4.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
TE	563.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	3200.0 ms
TE	563.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	Restore
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spcR
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear
Bandwidth	744 Hz/Px
Echo Spacing	3.52 ms
Turbo Factor	314
Echo Train Duration	1109 ms

Sequence - Part 2

Introduction	On
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	30 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\vessel_scout_head

TA: 22 sec Coil Selection: Auto Voxel Size: 1.2x1.2x5.0 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Geometry - AutoAlign

Slab Group	1
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A2.2 F32.9
L	0.0 mm
A	2.2 mm
F	32.9 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1
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Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\vessel_scout_head

TA: 22 sec Coil Selection: Auto Voxel Size: 1.2x1.2x5.0 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Geometry - AutoAlign

Slab Group	1
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A2.2 F32.9
L	0.0 mm
A	2.2 mm
F	32.9 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1
----------------	---

Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\vessel_scout_head

TA: 22 sec Coil Selection: Auto Voxel Size: 1.2x1.2x5.0 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Geometry - AutoAlign

Slab Group	1
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A2.2 F32.9
L	0.0 mm
A	2.2 mm
F	32.9 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1
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Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\vessel_scout_head

TA: 22 sec Coil Selection: Auto Voxel Size: 1.2x1.2x5.0 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Geometry - AutoAlign

Slab Group	1
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A2.2 F32.9
L	0.0 mm
A	2.2 mm
F	32.9 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1
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Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\vessel_scout_head

TA: 22 sec Coil Selection: Auto Voxel Size: 1.2x1.2x5.0 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Geometry - AutoAlign

Slab Group	1
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A2.2 F32.9
L	0.0 mm
A	2.2 mm
F	32.9 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1
----------------	---

Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3

TA: 2:40 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
R	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3

TA: 2:40 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
R	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3

TA: 2:40 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
R	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3

TA: 2:40 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
R	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3

TA: 2:40 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
R	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast pocs

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Read & Slice
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast pocs

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Read & Slice
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast pocs

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Read & Slice
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast pocs

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Read & Slice
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast pocs

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Read & Slice
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\asl_3d_tra_p2_iso_3mm_highres

TA: 9:51 min Coil Selection: Auto Voxel Size: 1.5×1.5×3.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
TE	20.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	5600.0 ms
TE	20.00 ms
Flip Angle	180 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Multiple Series	Off
Delay in TR	0.00 ms
Reordering	Centric

Contrast - ASL

Perfusion Mode	PCASL
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Contrast - ASL

Suppression	Gray-White
Labeling Duration	1800 ms
Postlabeling Delay	1800 ms
Delay Array Size	1

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	64
Phase Resolution	96 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	8
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Prescan

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8

Geometry - AutoAlign

Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Saturation

Special Saturation	Parallel F
Gap	35.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Rotation	-175.40 deg
A >> P	192 mm
R >> L	192 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5600.0 ms
Segments	5
Concatenations	1

Sequence - Part 1

Sequence Name	tgse
Dimension	3D
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Centric
Bandwidth	2232 Hz/Px
Echo Spacing	0.53 ms
Turbo Factor	10
Segments	5
EPI Factor	31

Sequence - Part 2

Introduction	Off
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\asl_3d_tra_p2_iso_3mm_highres

TA: 9:51 min Coil Selection: Auto Voxel Size: 1.5×1.5×3.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
TE	20.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	5600.0 ms
TE	20.00 ms
Flip Angle	180 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Multiple Series	Off
Delay in TR	0.00 ms
Reordering	Centric

Contrast - ASL

Perfusion Mode	PCASL
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Contrast - ASL

Suppression	Gray-White
Labeling Duration	1800 ms
Postlabeling Delay	1800 ms
Delay Array Size	1

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	64
Phase Resolution	96 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	8
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Prescan

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8

Geometry - AutoAlign

Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Saturation

Special Saturation	Parallel F
Gap	35.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Rotation	-175.40 deg
A >> P	192 mm
R >> L	192 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5600.0 ms
Segments	5
Concatenations	1

Sequence - Part 1

Sequence Name	tgse
Dimension	3D
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Centric
Bandwidth	2232 Hz/Px
Echo Spacing	0.53 ms
Turbo Factor	10
Segments	5
EPI Factor	31

Sequence - Part 2

Introduction	Off
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\Localizer

TA: 9 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	40.0 ms
TE	3.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	40.0 ms
TE	3.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	15 deg
Fat-Water Contrast	Standard

Contrast - Common

Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	75 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal

Geometry - Common

Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	40.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
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System - Adjustments

B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	40.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	75 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off

Inline - Subtraction

Save Original Images	On
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Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	15 deg
Measurements	1
Contrasts	1
TE	3.00 ms
TR	40.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\AAHScout

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Resolution - Common

Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	1.37 ms
TR	3.15 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\Localizer_aligned

TA: 21 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	104.0 ms
TE	3.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	104.0 ms
TE	3.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	15 deg
Fat-Water Contrast	Standard

Contrast - Common

Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	75 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8

Geometry - Common

Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	104.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	2
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
Slice Group	3
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
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System - Adjustments

B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	104.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	75 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off

Inline - Subtraction

Save Original Images	On
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Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	15 deg
Measurements	1
Contrasts	1
TE	3.00 ms
TR	104.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\SpinEchoFieldMap_AP

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0x2.0x2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	8036.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8036.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	On
EPI Factor	104

Sequence - Part 2

Introduction	Off
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Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\SpinEchoFieldMap_PA

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0x2.0x2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	8036.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8036.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	On
EPI Factor	104

Sequence - Part 2

Introduction	Off
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Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T1w_MPR

TA: 6:54 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
TE	2.22 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	2500.0 ms
TE	2.22 ms
Magn. Preparation	Non-sel. IR
TI	1000 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain

Geometry - AutoAlign

Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.2 A13.8 H12.4 mm
! Orientation	T > C-8.9 > S-0.5
! Rotation	4.64 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1000 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.22 ms
TR	2500.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear

Sequence - Part 1

Bandwidth	220 Hz/Px
Echo Spacing	7.54 ms
Asymmetric Echo	Allowed
Turbo Factor	256

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T2w_SPC

TA: 5:57 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
TE	561.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	3200.0 ms
TE	561.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm

Geometry - AutoAlign

H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.2 A13.8 H12.4 mm
! Orientation	T > C-8.9 > S-0.5
! Rotation	4.64 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	744 Hz/Px
Echo Spacing	3.98 ms
Turbo Factor	314
Echo Train Duration	1182 ms

Sequence - Part 2

Introduction	On
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCD_dMRI_DistortMAP_AP

TA: 30 sec Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	3
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	50 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off
Disable freq. update	Off

Sequence - Special

Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\rfMRI_REST_AP

TA: 6:40 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
TE	37.00 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain

Contrast - Common

TR	800.0 ms
TE	37.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	42 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	488
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
R	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.2 A13.8 H12.4 mm
Orientation	T > C-8.9 > S-0.5
Rotation	4.64 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	Ext./Trigger
Average Cycle	No Signal ms
Acquisition Window	800 ms
Trigger Pulse	1
Trigger Delay	0 ms
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	8
Phases	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	488
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	6600 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\CMRR_ABCDd102_81sl_89_3800_PA

TA: 6:47 min Coil Selection: Auto Voxel Size: 1.7×1.7×1.7 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
TE	89.20 ms
Multi-band accel. factor	3
AutoAlign	Head > Brain

Contrast - Common

TR	3800.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Standard
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.7 mm
Base Resolution	140
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	81
Distance Factor	0 %
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	1.7 mm
TR	3800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	3

Geometry - AutoAlign

Slice Group	1
Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.5 A5.9 H13.4 mm
Orientation	T > C-17.9 > S-1.3
Rotation	4.49 deg
A >> P	240 mm
R >> L	240 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3800.0 ms
Multi-band accel. factor	3

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	3
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Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	On
FA Maps	On
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	20
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	On
EPI Factor	140

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off

Sequence - Special

Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\T1w_MPR

TA: 6:54 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
TE	2.22 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	2500.0 ms
TE	2.22 ms
Magn. Preparation	Non-sel. IR
TI	1000 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain

Geometry - AutoAlign

Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.2 A13.8 H12.4 mm
! Orientation	T > C-8.9 > S-0.5
! Rotation	4.64 deg
! A >> P	208 mm
! R >> L	208 mm
! F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1000 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.22 ms
TR	2500.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear

Sequence - Part 1

Bandwidth	220 Hz/Px
Echo Spacing	7.54 ms
Asymmetric Echo	Allowed
Turbo Factor	256

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\t2_space_sag_cs4_iso

TA: 2:22 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.8 mm³ Acc:: 4.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
TE	563.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	3200.0 ms
TE	563.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	Restore
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	93.8 %
Slice Thickness	0.80 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spcR
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Performance
Flow Compensation	None
Reordering	Linear
Bandwidth	744 Hz/Px
Echo Spacing	3.52 ms
Turbo Factor	314
Echo Train Duration	1109 ms

Sequence - Part 2

Introduction	On
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	30 s

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\vessel_scout_head

TA: 22 sec Coil Selection: Auto Voxel Size: 1.2x1.2x5.0 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Geometry - AutoAlign

Slab Group	1
Position	R0.8 A11.2 F26.5 mm
Orientation	S > C4.7 > T-0.2
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 A2.2 F32.9
L	0.0 mm
A	2.2 mm
F	32.9 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1
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Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3

TA: 2:40 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.6 F5.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
R	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\tof_cs_acc10.3 fast pocs

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.6 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms
TE	3.00 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.08 ms
TE	3.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Read & Slice
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.56 mm
TR	18.08 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	R0.9 A11.3 F3.5 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.6 F10.1
L	0.0 mm
P	0.6 mm
F	10.1 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.08 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.00 ms
TR	18.08 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.38 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Mair test\\asl_3d_tra_p2_iso_3mm_highres

TA: 9:51 min Coil Selection: Auto Voxel Size: 1.5×1.5×3.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
TE	20.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	5600.0 ms
TE	20.00 ms
Flip Angle	180 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Multiple Series	Off
Delay in TR	0.00 ms
Reordering	Centric

Contrast - ASL

Perfusion Mode	PCASL
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Contrast - ASL

Suppression	Gray-White
Labeling Duration	1800 ms
Postlabeling Delay	1800 ms
Delay Array Size	1

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	64
Phase Resolution	96 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	8
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Prescan

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8

Geometry - AutoAlign

Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Saturation

Special Saturation	Parallel F
Gap	35.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Rotation	-175.40 deg
A >> P	192 mm
R >> L	192 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5600.0 ms
Segments	5
Concatenations	1

Sequence - Part 1

Sequence Name	tgse
Dimension	3D
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Centric
Bandwidth	2232 Hz/Px
Echo Spacing	0.53 ms
Turbo Factor	10
Segments	5
EPI Factor	31

Sequence - Part 2

Introduction	Off
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Sequence - Assistant

SAR Assistant	Off
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